TOWARDS CRITICAL PHENOMENA IN TWISTING AXISYMMETRIC VACUUM SPACETIMES

A Dissertation Submitted
in Partial Fulfilment of the Requirements
for the Degree of

MASTER OF SCIENCE

in **Physics**

by

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to

SCHOOL OF PHYSICS
INDIAN INSTITUTE OF SCIENCE EDUCATION AND RESEARCH THIRUVANANTHAPURAM
INDIA - 695 551

April 2026

CERTIFICATE

This is to certify that this dissertation entitled **"Towards Critical Phenomena in Twisting Axisymmetric Vacuum Spacetimes"** submitted by **Ashique T Nizar (Reg. No.: IMS21039)** towards the partial requirement of **Master of Science** in **Physics**, has been duly examined by the thesis committee appointed by the institute. The committee deems the candidate's work satisfactory and recommends that the report be accepted.

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DECLARATION

I, **Ashique T Nizar (Reg. No.: IMS21039)**, hereby declare that this dissertation entitled **"Towards Critical Phenomena in Twisting Axisymmetric Vacuum Spacetimes"**, submitted to Indian Institute of Science Education and Research, Thiruvananthapuram, towards the partial requirement of **Master of Science** in **Physics**, is a bona fide record of original work carried out by me under the supervision of **Dr.Soumen Basak**.

This dissertation has never been submitted in part or in full for a degree, diploma or fellowship to this or any other university before. I duly acknowledge all external contributions, statements, datasets or results used, and have listed their sources with adequate detail in the bibliography.

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In my capacity as the project supervisor for the aforementioned candidate, I confirm that the above statements by the candidate are true to the best of my knowledge.

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ABSTRACT

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Thesis Title: **Towards Critical Phenomena in Twisting Axisymmetric Vacuum Spacetimes**
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Date of thesis submission: **15 April 2026**

Critical phenomena in gravitational collapse constitute one of the earliest major results in numerical relativity, revealing universal behavior at the threshold of black hole formation. In this work, we investigate the emergence of such phenomena in axisymmetric vacuum spacetimes with twist (which encodes the rotational degree of freedom of the spacetime). By constructing and evolving one-parameter families of twisting Brill wave initial data, we study the threshold between dispersion and black hole formation. Near the critical threshold, we analyze scaling behavior using curvature invariants and examine indications of universality and self-similarity. Our results show evidence of power-law scaling and suggest the presence of echo-like features, although higher resolution is required for conclusive identification. We also identify key numerical challenges, including gauge effects, coordinate pathologies, and limitations in horizon-finding techniques, which become significant near criticality. This work provides insight into critical phenomena in vacuum spacetimes beyond spherical symmetry and highlights the role of twist in gravitational collapse.

Keywords:

Numerical Relativity, Apparent Horizons, Singularity

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Chapter 1

Introduction and Background

General Relativity (GR) provides a profound description of gravity as the dynamics of spacetime geometry, governed by a set of highly nonlinear Partial Differential Equations (PDEs). While many exact solutions to Einstein's equations are known, analytic methods often limit the wide variety of relativistic systems that could be understood using the framework of GR. This necessitates the use of numerical methods, leading to the field of Numerical Relativity (NR), in which spacetime dynamics are studied computationally.

One of the most intriguing phenomena uncovered through numerical studies is critical behavior in gravitational collapse. When evolving a one-parameter family of initial data, the system exhibits a threshold separating dispersion from Black Hole (BH) formation. Near this threshold, the spacetime displays remarkable universal features such as power-law scaling, universality of critical exponents, and self-similarity. These discoveries, first made in spherically symmetric systems, revealed deep connections between gravity, nonlinear dynamics, and critical phenomena.

However, much of our current understanding is restricted to highly symmetric settings. In particular, spherical symmetry eliminates gravitational wave degrees of freedom, limiting the richness of the dynamics. To explore genuinely gravitational critical phenomena, it is essential to move beyond spherical symmetry into axisymmetric or fully three-dimensional spacetimes. In vacuum, this becomes especially significant, as nontrivial dynamics arise purely from the nonlinear structure of Einstein's equations rather than matter fields.

In this work, we investigate critical phenomena in axisymmetric vacuum spacetimes using Brill wave initial data, with a particular focus on the role of twist, which introduces additional geometric degrees of freedom. By constructing and evolving one-parameter families of twist-included initial data, we aim to study how these additional structures influence the nature of the critical solution, its universality, and the emergence of self-similar behavior. In this chapter, we provide a concise overview of the key background concepts relevant to this work.

1.1 Numerical Relativity

Numerical Relativity can be defined as the science of solving Einstein's equations using numerical methods in such a way that the discretized solutions agree with the continuum solutions at the limit of infinite resolution. They are a set of coupled 4-dimensional PDEs that connect the dynamics between matter (by stress energy tensor ($T_{\mu\nu}$)) and the spacetime curvature (Through the Einstein tensor $G_{\mu\nu}$).

$$G_{\mu\nu} = 8\pi T_{\mu\nu}$$

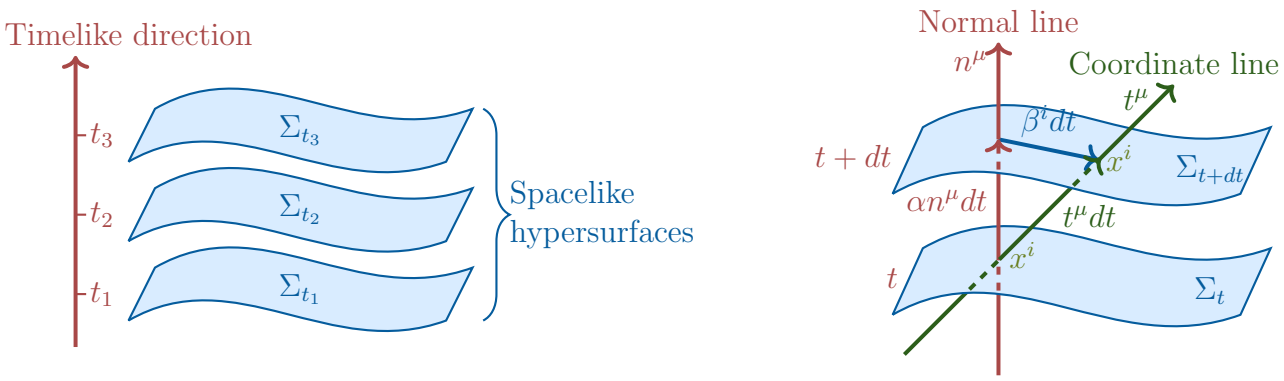


Figure 1.1: (a) Spacelike hypersurfaces sliced along the timelike direction. (b) Relation between slices showing lapse α and shift β^i. [1]

But its tensorial structure prevents us from approaching them as usual time-evolution PDE problems, where we see equations like $\partial_t u = f(x^i, \partial_{x^i} u)$, where x^i are the spatial coordinates and t is the time coordinate. So in order to cast Einstein's equations into a form suitable for time evolution, we assume that the spacetime is globally hyperbolic. This guarantees the existence of a foliation of the spacetime manifold ($\mathcal{M}$) into space-like Cauchy hypersurfaces Σ_t, which are the level sets of a scalar time function $t : \mathcal{M} \to \mathbb{R}$. Each slice admits a unique future-directed timelike unit normal vector [1]

$$n^a = -\alpha \nabla^a t, \qquad \alpha^{-2} = -g_{ab} \nabla^a t \nabla^b t,$$

where α is the lapse function, g_{ab} is the spacetime metric, and ∇_a is the covariant derivative compatible with g_{ab}. This construction allows the decomposition of Einstein's equations into evolution and constraint equations within the $3+1$ formalism. In a coordinate basis adapted to the foliation (∂_t, ∂_i), the vectors ∂_i lie within the hypersurfaces Σ_t, while $(\partial_t)^a$ need not be aligned with the normal vector n^a. Instead, ∂_t generally maps a point on one slice to a point with the same coordinate label on the next slice, whereas n^a connects orthogonal (normal) directions across slices. The difference is encoded in the shift vector β^i through the standard decomposition $\partial_t^a = \alpha n^a + \beta^i \partial_i^a$, which parametrizes how spatial coordinates shift from one hypersurface to the next.

By using the orthogonal projection operator to the hypersurface $\perp_{ab} = \delta_{ab} + n_a n_b$, we can project all the relevant quantities that we require, such as the metric, Riemann curvature tensor, etc. This projected part of g_{ab} is called the spatial metric γ_{ab}, and D is the derivative associated with this metric. We define extrinsic curvature K_{ab} to be $-\gamma_a^c \nabla_c n_b$ which in simple terms give the information of the time derivative of the spatial metric by the relation $K_{ab} = -\mathcal{L}_n \gamma_{ab}$ where $\mathcal{L}$ is the lie derivative with respect to a vector field. Using all these, we can decompose Einstein's equations into a set of constraint equations and evolution equations as given below.

$$R + K^2 - K_{ab}K^{ab} = 16\pi\rho \qquad D_b K^{ab} - D^a K = 8\pi j^a, \tag{1.1}$$

which correspond to Hamiltonian and momentum constraints respectively, R corresponds to

[1] We use abstract index notation following Wald [15] or Hawking and Ellis [7].

Ricci scalar associated with the γ_{ab}, $j^a = -\perp T_{ab}n^b$, $s_{ab} = \perp T_{ab}$, $\rho = T_{ab}n^a n^b$ [2]

$$\partial_t \gamma_{ij} = -2\alpha K_{ij} + \mathcal{L}_\beta \gamma_{ij},$$

$$\partial_t K_{ij} = -D_i D_j \alpha + \alpha(R_{ij} - 2K_{ik}K^k{}_j + KK_{ij}) + 4\pi\alpha[\,\gamma_{ij}(s-\rho) - 2s_{ij}] + \mathcal{L}_\beta K_{ij}.$$ [3]

Now we can see that the GR equations have been formulated in a way that is ideal for time evolution using standard PDE techniques (Since all the elements in the RHS are spatial). However, more modern formulations such as BSSN, Z4C, and related systems include constraint-damping mechanisms that improve numerical stability, since in a discretized scheme the constraints cannot be satisfied exactly. It is also well known that the constraint subsystem is closed, meaning that the time derivatives of the constraints depend only on the constraints themselves. Consequently, if the constraint equations are solved initially, they remain satisfied for a finite time during the evolution. For further details, see [8]. So we take solutions of the constraint equations, which often amount to solving elliptic PDE's on the initial slice (called **Initial Data**) and then proceed to evolve it in time using the evolution equations.

1.2 Critical Phenomena in Gravitational Collapse

Instead of viewing GR as a field theory, a dynamical systems perspective helps us see a widely underappreciated nuance of the theory: GR is essentially a set of PDEs at its core, which enables us to view it as such. Essentially, the dynamic system's phase space will be a set of functions of the coordinates, namely the spatial 3-metric, the extrinsic curvature, and the matter variables we choose. So as we evolve in time, these points in the phase space will evolve from one point to another while satisfying the Einstein equations as well as the matter equations that we set. Until now its pretty standard. To see the nuance, we take a one-parameter family of smooth initial data (Solving the Einstein constraint equations (Eq:1.1) and the matter equations), we observe that they usually reach two types of end points in phase space, either disperse the mass-energy to infinity (known as **subcritical** initial data) or collapse to a black hole (known as **supercritical** initial data).

But the main interest lies in the spacetimes that sit *in the critical regime*, between the two end-states of dispersion and black-hole(BH) formation. In his breakthrough numerical study, Choptuik [4], using spherical symmetry with a minimally coupled, massless scalar field as the source of stress–energy, discovered that this intermediate regime exhibits three remarkable and universal features:

- **Power-law mass scaling:**[4]
  On the supercritical side, the mass of the collapsed BH obeys a scaling law of the form

[2] $\perp$ means using projection operator on every free index

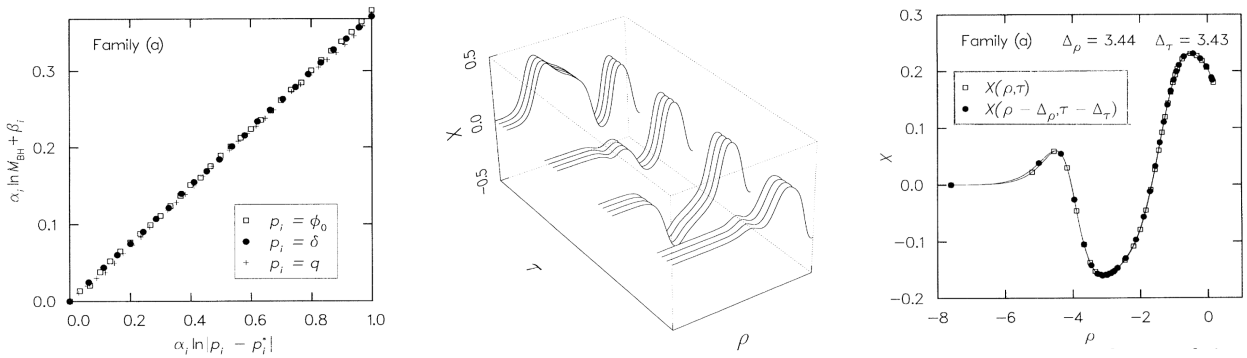
[3] where i,j $\rightarrow$1,2,3 (Spatial)

[4] Later Garfinkle and Comer [5] found out that according to proper dimensionality, the maximum of curvature invariants from the subcritical side also follow similar scaling laws (Kretschmann Scalar maximum $K_{max} = (p - p^*)^{-4\eta}$) with the same critical exponents.

$M_{\mathrm{BH}} \propto (p - p_*)^\eta$ where p is the parameter controlling a one-parameter family of initial data, p_* is the critical value separating the two regimes, and η is the *critical exponent*. Choptuik found the critical parameter with up to 15-digit precision. See Fig 1.2(a)

- **Universality of the exponent:**
  The exponent η is **independent of the choice of initial data family**. Different families with different shapes, widths, or profiles all yield the same value of η, depending only on the matter model (e.g., scalar field, Yang–Mills field, etc.). See Fig 1.2(a)

- **Self-similarity and universality of the critical solution:**
  The critical solution itself exhibits a form of **self-similarity** (either continuous or discrete), meaning that the spacetime repeats its structure on smaller and smaller spatiotemporal scales based on how near the parameter is to the critical parameter. Additionally, this critical solution is universal with respect to initial data. See Figs 1.2(c) 1.2b



(a) We can observe here the theorized power-law mass scaling in the supercritical data. Different dots signify different families of initial data, and they all agree with the exponent (Power law mass scaling and universality of exponent)

(b) Here, each set of 4 curves contains near-critical solutions of different families of initial data. As it evolves in time, we can see that they even have the same profile (Universality of critical solution)

(c) In the critical solution, the same structure (echoes) reappears under rescaling. The two different dots correspond to the same field at different times, and zooming in by a factor of e^Δ reveals the same repeating pattern, and this constant is family independent known as the echoing constant (Self-similarity).

Figure 1.2: Critical Phenomenon from [4]

The choice of spherical symmetry here was crucial: it reduced the full $(3+1)$-dimensional PDE system to an effective $(1+1)$-dimensional problem (in suitable coordinates, the fields depend only on time and radius). This simplification dramatically lowers the computational cost compared with a full three-dimensional evolution, while still capturing highly nontrivial gravitational dynamics. Following Choptuik's discovery, many researchers studied and characterized critical phenomena in a wide variety of matter models like scalar fields, perfect fluids, Yang-Mills fields, and scalar electrodynamics, etc., mostly within spherical symmetry, taking advantage of the same reduction in dimensionality and computational expense [6].

To further understand the structure of the critical solution space of Einstein's equations, subsequent studies have explored departures from spherical symmetry. In this context, one of the

most important and intriguing cases is that of vacuum spacetimes, i.e., gravitational collapse without matter fields. In spherical symmetry, vacuum solutions are trivial due to Birkhoff's theorem, which implies the absence of dynamical degrees of freedom. Therefore, going beyond spherical symmetry is essential to probe genuinely gravitational critical phenomena in vacuum, where the dynamics arise purely from the non-linear structure of Einstein's equations.

In this regard, axisymmetric vacuum spacetimes form the next natural class of models, as they admit nontrivial gravitational wave dynamics absent in spherical symmetry, while still offering a significant reduction in computational complexity compared to fully three-dimensional simulations. In this class, the two main candidates are Brill waves and Teukolsky waves. In this work, we focus on Brill waves because they are comparatively simpler to implement.

1.2.1 Brill waves

Brill waves are axisymmetric, vacuum gravitational-wave initial data constructed on a moment-of-time-symmetry slice ($K_{ij} = 0$), providing a practical prescription for generating axisymmetric vacuum spacetimes, which is obtained by solving the constraint equation in axisymmetric vacuum. These are then evolved in time to learn curvature dynamics of pure non-linear gravity rather than matter. Brill considered this ansatz for the spatial axisymmetric metric in cylindrical coordinates.

$$\gamma_{ij}(\rho, z, \phi) = \gamma_{ij}(\rho, z) = \psi^4 \begin{pmatrix} e^{2q(\rho,z)} & 0 & 0 \\ 0 & e^{2q(\rho,z)} & 0 \\ 0 & 0 & \rho^2 \end{pmatrix}, \qquad (i, j) = (\rho, z, \phi).$$

When $q(\rho, z) = 0$, the seed metric on the (ρ, z)–plane is flat, so the Brill 3-metric reduces to a conformally flat geometry. Thus $q(\rho, z)$ acts as a *seed function* that introduces nontrivial gravitational content into the initial data. At the same time, the conformal factor ψ is fixed by the Hamiltonian constraint, which becomes an elliptic equation [5] to solve, making this metric constraint satisfying. When $\gamma_{ij} = \psi^4 \tilde{\gamma}_{ij}$ then the Hamiltonian constraint becomes , $8\tilde{D}^2\psi = \tilde{R}\psi$ where $\tilde{R}$ is the Ricci scalar of the conformal metric and $\tilde{D}$ is the derivative operator compatible to the conformal metric. This, in our case, becomes a linear differential equation of the form in Eq.1.2. This is the main advantage of using this conformal formalism: we just need to solve this differential equation to obtain the initial data. And since $K_{ij} = 0$, the momentum constraint will be trivially satisfied (See equation 1.1).

$$\tilde{D}^2\psi = -\frac{1}{4}\left(\frac{\partial^2 q}{\partial \rho^2} + \frac{\partial^2 q}{\partial z^2}\right)\psi \tag{1.2}$$

A spacetime is axisymmetric if it admits globally a spacelike Killing vector field $\xi^a = (\partial_\phi)^a$ whose integral curves are closed, reflecting the periodicity of ϕ. In the untwisted Brill metric, the components $g_{i\phi}/\gamma_{i\phi}$ vanish, so this Killing vector is **twist-free**. The twist measures the failure of ξ^a to be hypersurface orthogonal, and is encoded in the 1-form $\omega_a = \epsilon_{abcd}\,\xi^b \nabla^c \xi^d$

[5] With York's conformal decomposition method

Motivation

Critical collapse in twist-free axisymmetric geometries, both in vacuum and non-vacuum settings, has revealed significant departures from the standard critical phenomena observed in spherical symmetry, including bifurcation of the collapse point and loss of universality in scaling laws [3]. These deviations suggest that the additional gravitational degrees of freedom present in axisymmetry, namely, gravitational waves, might play a crucial role, as opposed to purely matter-driven dynamics.

But we also see that the introduction of rotating matter (with angular momentum) restores some of the textbook critical behavior that is lost in the previous case (only in non-vacuum cases), such as a single center of collapse and universal scaling laws [12].[6]. A possible explanation for the restoration of standard critical behavior in the presence of angular momentum is that, in axisymmetry, there is no corresponding vacuum threshold solution with nonzero angular momentum. This is because with regular vacuum initial data (without singularities/BH), the data can't have or evolve to have angular momentum [12]. As a result, the dynamics is not influenced by competing vacuum critical solutions, allowing the system to exhibit cleaner, more universal behavior. In contrast, in the absence of angular momentum, vacuum axisymmetric thresholds may introduce competing dynamics that lead to deviations from the standard picture. But the absence of angular momentum doesn't necessarily imply the absence of twist.

Hence, to assess the hypothesis, it is essential to investigate whether a distinct twisting vacuum critical solution exists. If such a solution is present and lies nearby in solution space, it could influence or compete with matter-driven collapse, potentially modifying or suppressing universal behavior even in the case with angular momentum in non-vacuum

This motivates our investigation: to introduce twist into the Brill metric (adding two new degrees of freedom) and examine how twist influences the critical solution and its universality. Now we introduce the metric for Brill-type initial data with twist included.

$$\gamma_{ij}(\rho,z,\phi)=\gamma_{ij}(\rho,z)=\psi^4\begin{pmatrix} e^{2q(\rho,z)}+f^2(\rho,z) & f(\rho,z)h(\rho,z) & \rho f(\rho,z) \\ f(\rho,z)h(\rho,z) & e^{2q(\rho,z)}+h(\rho,z)^2 & \rho h(\rho,z) \\ \rho f(\rho,z) & \rho h(\rho,z) & \rho^2 \end{pmatrix}, (i,j)=(\rho,z,\phi). \tag{1.3}$$

where the function *f(ρ,z)* and *h(ρ,z)* parameterises the twist of the spacelike killing vector. This ansatz is chosen because it cleanly splits into a twist-free Brill metric and a twist part when the line element is written. The new differential equation to solve for the conformal factor becomes

$$8\tilde{D}^2\psi=\left(-\frac{e^{-2q}(\rho(\frac{\partial f}{\partial z}-\frac{\partial h}{\partial \rho}))+h)^2}{2\rho^2}-2\left(\frac{\partial^2 q}{\partial \rho^2}+\frac{\partial^2 q}{\partial z^2}\right)\right)\psi \tag{1.4}$$

[6]When there is angular momentum, there will be twist but not vice versa

1.3 Horizons

The main tool we use to classify spacetimes as subcritical or supercritical is to probe for horizons, specifically Apparent Horizons (AHs), in the evolved slice. The main advantage of AH is that, unlike event horizons, they are not *teleological*, i.e., we don't need the information about the entire future development of the spacetime to get the information about AHs. In numerically computed dynamical spacetimes like ours, we might not be able to evolve them indefinitely to classify them, since the code will crash after some further evolution. Hence, AHs are the right tool to understand them.

But what is an AH and why does it relate to the notion of a singular spacetime?

Essentially, an AH can be defined as the outermost Marginally Outer Trapped Surface (MOTS), which is a smooth, closed 2-dimensional submanifold $\mathcal{S}$ in the spacetime slice that satisfies the condition that the expansion of the outgoing null geodesics vanishes at each point in $\mathcal{S}$. The expansion is defined as

$$\Theta = h^{ab}\nabla_a\zeta_b \qquad \text{where } h^{ab} = g^{ab} - s^a s^b + n^a n^b$$

h^{ab} is the induced metric of the 2-dimensional submanifold, s^a is its normal vector and ζ is the tangent vector along the outgoing null geodesic . Solving this, we get the equation of the form,

$$\Theta = -K + D_i s^i + s^i s^j K_{ij} \tag{1.5}$$

Hence, our aim is to find surfaces satisfying ($\Theta = 0$), and we identify the apparent horizon (AH) as the **outermost marginally outer trapped surface (MOTS)**.

The importance of apparent horizons is closely tied to the singularity theorems of general relativity. These theorems state that, under suitable energy conditions (such as the *null energy condition*, which is satisfied in vacuum since ($T_{ab} = 0$)) and global hyperbolicity (satisfied since the initial data is constraint satisfying), the presence of a trapped surface implies that the spacetime is geodesically incomplete, i.e., contains a singularity.

Thus, the formation of a MOTS or apparent horizon serves as a strong indicator of BH formation in the spacetime's future development. This provides a practical and geometric criterion for classifying spacetimes in our analysis.

Chapter 2

Methodology

2.1 Construction of 1-parameter family of Initial Data

For solving the constraint equation on the initial $t = 0$ slice, we employ a pseudospectral method. Since the equation is a linear PDE in two spatial dimensions and the data are smooth, the spectral approach provides exponential convergence as the number of collocation points is increased. In contrast, obtaining a closed-form solution is highly impractical: the variable coefficients introduced by general choices of $q(\rho, z)$, $f(\rho, z)$, and $h(\rho, z)$ prevent separation of variables and lead to nonstandard equations. Moreover, pursuing an analytic solution would restrict our freedom to modify the ansatz.

For the first step, inspecting the most general axisymmetric and regular metric ansatz [13] shows that $f(\rho, z)$ must be an even function of ρ and $h(\rho, z)$ must be an odd function of ρ (at origin). In addition, the metric must approach the flat metric at spatial infinity to ensure a well-defined notion of mass for any BH that forms and to keep the analysis confined to the isolated system under consideration [1]. While evolving, it's always better to have some form of parity with respect to z as well, since it will decrease the cost of evolutions (We can reflect with respect to the z axis)(From 1.4). Hence, considering all these factors, we take the ansatz

$$f(\rho, z) = \text{twp1}\, \rho^2 z\, e^{-(\rho^2+z^2)} \quad h(\rho, z) \quad = \text{twp2}\, \rho^3 e^{-(\rho^2+z^2)}. \tag{2.1}$$

where twp1 and twp2 are parameters that we would like to characterize the critical phenomenon by constructing some 1-parameter families. As for the 1-parameter family, for now we choose twp1 = twp2 = p, where p is the parameter we vary and tune to the threshold of collapse.

2.2 Time Evolutions of Twist-included Brill wave Initial Data

With the new twist included Brill wave initial data that we constructed, we evolved them in `BAMPS` NR framework [9]. `BAMPS` is a pseudospectral code based on the method of lines, using Chebyshev expansions on Gauss–Lobatto collocation points. The computational domain is constructed using multi-patch cubed-sphere grids, with subpatches enabling Adaptive Mesh Refinement(AMR) and parallelization. Additionally, due to the axisymmetric nature of the problem at hand, we use the cartoon method to simplify the computational costs.

[1] For asymptotic flatness

Cartoon Method [2]

In general, performing full 3-D simulations in NR is computationally expensive due to the problem's high dimensionality. However, when the system possesses axisymmetry, one can exploit this symmetry to significantly reduce computational cost without losing essential physical information.

The cartoon method is a technique for evolving axisymmetric spacetimes using a reduced computational domain while still employing a Cartesian grid. Instead of evolving the full three-dimensional domain, the evolution is restricted to a thin slab around the symmetry plane (typically the $y = 0$ meridional plane), consisting of only a few grid points in the symmetry direction.

The key idea is that derivatives in the symmetry direction (e.g., the y-direction) can be reconstructed using the assumption of axisymmetry. This is achieved by performing suitable rotations of the evolved variables around the symmetry axis. For a tensor field T, its value at off-plane points required for finite difference stencils is obtained by rotating the on-plane data using the Killing symmetry associated with rotations about the axis. Hence, essentially our computational domain for the whole problem is the **Cartesian xz (y=0)** plane.

Harmonic Damped Wave Gauge (HDWG)

Harmonic Damped Wave Gauge (HDWG) is obtained by modifying the harmonic gauge condition ($\nabla_a \nabla^a x^b = 0$) with a damping term in the RHS, turning the coordinate condition into a damped wave equation ($\nabla_a \nabla^a x^b = H^b$). This suppresses unphysical gauge modes and stabilizes the coordinate evolution during numerical simulations. The gauge source function/RHS for it is of the form.

$$H_a = \eta_L \log\left(\frac{\gamma^p}{\alpha}\right) n_a - \eta_s \alpha^{-1} g_{ai} \beta^i$$

where η_s, η_L and p are constants. In black-hole spacetimes, particularly near horizon formation, one often observes a rapid growth of the spatial metric determinant γ due to coordinate stretching. The first term in H_a, through the logarithmic factor $\log(\gamma^p/\alpha)$, acts to counter this growth by driving a collapse of the lapse α, thereby controlling the increase in spatial volume elements. The second term damps the shift vector β^i, suppressing distortions in the spatial coordinates. For more details, refer [11]

Together, these terms act to control gauge dynamics, preventing the development of large coordinate distortions and stabilizing the evolution near strong-field regions. Since η_L and η_s set the time scale where these damping acts, a bit of tuning to find the right parameters is required for proper gauge evolutions, which becomes essential in proper evolutions.

2.3 Horizon Finding

In our axisymmetric setting, using a spherical parameterization to locate surfaces satisfying the vanishing expansion condition in Eq.1.5, the problem reduces to a one-dimensional ordinary

differential equation in the polar angle. To solve them, we use the `AHLoc3D` tool with a parabolic flow method [14]. Essentially, it makes use of guess surfaces in such a way that the outermost MOTS is inside the surface, calculates the expansion at a number of points based on the resolution and the "flows" towards lesser and lesser expansion values based on a pseudo time stepping (λ), ie $\partial_\lambda x^i = -\Theta s^i$ where s^i is the normal vector to the surface and x^i are the coordinates of points in the guess surface.

Additionally, in the case of `AHLoc3D`, the spherical parameterization used, which is $r = F(\theta)$, can only find ray-body/star-shaped horizons. This means that if we draw a ray from the center to the horizon, it should only touch the horizon once, i.e., the function should not be multivalued in r. This parameterization begins to break down even before the surface becomes strictly multivalued. In particular, when the surface develops regions where the radial direction becomes tangent to the horizon, the function $F(\theta)$ develops large gradients or vertical tangents. Numerically, this corresponds to a loss of regularity in the representation, and the horizon finder can fail to converge even though a smooth surface still exists.

Thus, the method is limited not only by true multivalueness, but also by configurations that approach it, in which the surface ceases to be well described as a function $r(\theta)$.

Despite this limitation, such a parameterization significantly simplifies the apparent horizon finding problem by reducing it to a one-dimensional equation in θ, making it computationally efficient and straightforward to automate. Consequently, it has become a practical and widely adopted approach in NR. Moreover, prior studies of vacuum critical collapse have not indicated

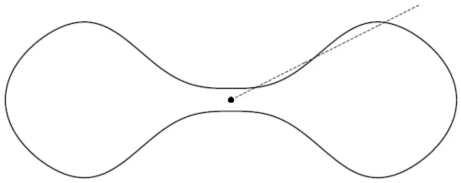


Figure 2.1: An example for a non-ray body horizon. Here, the r = F(θ) parameterization breaks, and `AHLoc3D` method won't be able to find this type of a horizon in the spacetime [14].

the necessity for more general surface representations, as the horizons encountered in those scenarios remain well approximated by star-shaped surfaces. But as we will discuss in the following sections, the inability to tune to the threshold led us to rethink this assumption as a hypothesis, which proved to be the correct approach.

We also use a complementary method based on a Cartesian parameterization with a shooting algorithm, which reduces to solving a set of coupled ODEs in axisymmetry. This removes the ray body limitation that we faced previously, but the method uses the Cartoon method as in Sec.2.2 since this search is limited on the y=0 plane and the surface becomes a curve which is parameterized as $X^i(\lambda) = (X(\lambda), 0, Z(\lambda))$

For this method, we integrate Eq.1.5 with this new parameterization, obtaining two coupled ODEs in X and Z. Along the symmetry axis, we begin by making initial guesses for the point at which the horizon intersects the z-axis. For each such guess, we integrate the coupled ODEs

to obtain a candidate curve $(X(\lambda), Z(\lambda))$. This defines a shooting problem, where we search for solutions that return to the axis, i.e., satisfy $X(\lambda_{\max}) = 0$ for some $\lambda_{\max} > 0$. In addition to this closure condition, regularity at the axis requires the tangent vector to be aligned with the radial direction in the meridional plane, which imposes the boundary condition $u^z(\lambda_{\max}) = 0$. We therefore select those curves that both return to the axis and satisfy this smoothness condition.

Finally, these candidate solutions are passed to a root-finding algorithm, such as `BRENTQ`, to determine the precise initial condition that yields a closed, smooth horizon. For further details, see [10].

General workflow for tuning to the threshold of collapse

1. Initial Data :- Using the framework that we made for solving the Initial Data PDE for arbitrary parameters, we construct the initial data for whatever parameters we need for tuning. For example, if we know the initial data is between [1,2] then we will construct 10 initial data like 1.0,1.1,1.2.....2.0
2. Evolve these initial data using `BAMPS` and analyze the results. If we observe dispersion on one parameter and horizon formation in the subsequent one, do the next level of search between these parameters and repeat the process
   - For subcriticality, we look at the Curvature invariant, Kretschmann Scalar

     $$K = {}^{(4)}R_{abcd}\,{}^{(4)}R^{abcd}$$

     where ${}^{(4)}R_{abcd}$ is the Riemann curvature tensor in 4D, observe its behaviour and check whether its dispersing towards spatial infinity (the end of the grid) and ensure there is no reflection at the boundaries which can be problematic.
   - For supercriticality, if the code crashes (either due to numerical error or singularity), we output the very expensive apparent horizon data from the previous checkpoint time for a short time and use that data to find whether horizons have formed or not.
3. If we get to a scenario where we can't determine it due to errors or coordinate artifacts, we adjust the relevant gauge parameters (in 2.2) and evolve them again until we can determine them properly.

Once the critical parameter has been determined to the desired precision (typically five to six decimal places), we investigate the behavior of the system in its vicinity. In particular, we focus on the universality of critical quantities and analyze the emergence of (discrete) self-similarity through the appearance of self-similar "echoes" in the relevant fields. In that regard, we try to measure the exponents from the maximum of curvature scalars from the subcritical side [5] rather than use the BH mass as an indicator (Measured from AH area). This choice is motivated by the ambiguity associated with determining when to measure the AH area, especially when comparing different evolutions. Additionally, AH is a slicing-dependent object, and since we adjust the gauges based on the numerical stability it provides to the simulations, it removes any basis for comparison.

Chapter 3

Observations, Results and Challenges

Critical phenomena in gravitational collapse are most clearly understood in spherical symmetry, where the approach to the critical regime is comparatively straightforward. In this case, the field equations reduce to a system of coupled one-dimensional PDEs, which can be solved with relatively high numerical accuracy. Indeed, in the pioneering work of Matthew Choptuik [4], the critical parameter was tuned to nearly machine precision, reaching approximately 15 decimal places.

However, once symmetry assumptions are relaxed, the problem becomes significantly more challenging. In the literature, studies of axisymmetric vacuum collapse typically achieve only about 5–7 digits of precision in tuning the critical parameter. This limitation arises from both the increased computational difficulty of solving multidimensional PDEs and the intrinsic nature of critical phenomena, which involve the development of increasingly fine spatial and temporal structures near the threshold of black hole formation.

For the family of initial data considered in this work, the best-tuned critical parameter is found to be $p^* = 4.110 \pm 0.003$ for positive values of p and $p^* = -4.110 \pm 0.003$ for negative values of p. Despite this comparatively modest level of tuning, we are able to extract a critical exponent of approximately $\gamma \approx 0.354 \pm 0.204$. This suggests that the scaling behavior is reasonably robust even without extremely fine-tuned initial data. In Fig. 3.1, we observe that two families (corresponding to positive and negative values of p, as described in Sec. 2.1) yield consistent estimates of the critical exponent.We obtain an R^2 value of 0.995 for the linear fit to the log–log plot of $K_{\max}$ versus $|p - p^*|$, indicating good agreement with an approximate scaling behaviour.

However, these results should be interpreted with caution. More precise tuning toward the critical threshold, along with a broader set of initial data families, is required before drawing stronger conclusions, particularly regarding universality. This is because at this level of tuning, the system may still be sufficiently far from the critical regime that the extracted exponent is unreliable. Previous studies suggest that insufficient fine-tuning can introduce systematic deviations in the inferred scaling behavior.

The quoted uncertainty in the critical parameter arises from the inability to unambiguously distinguish between subcritical and supercritical evolution within this narrow region of parameter space. In particular, near the threshold, small variations in the parameter lead to qualitatively different outcomes, and within our numerical resolution, this transition cannot be resolved more precisely. This limitation propagates into the determination of the scaling exponent, contributing to the relatively large uncertainty in (γ).

However, further refinement of the tuning toward the critical regime is required to investigate the critical behavior more reliably, including the possible presence of self-similar structures. For

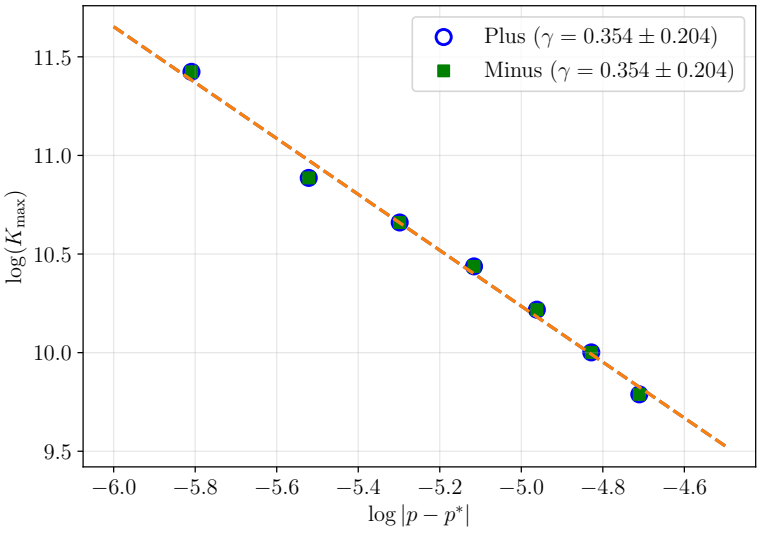


Figure 3.1: Log–log scaling of the maximum Kretschmann scalar $K_{\max}$ as a function of $|p - p^*|$ for both families. The fitted lines give consistent estimates of the critical exponent γ, which is to be trusted only after further tuning. Each point corresponds to an independent simulation worth ≈ 4000 core hours. The approximately linear behavior is consistent with the expected scaling $K_{\max} \sim (p^* - p)^{-4\gamma}$ near criticality.

example, as the parameter is tuned closer to the critical value, the evolution exhibits increasingly pronounced oscillatory features in the curvature, suggestive of the onset of echoing behavior as in Fig 3.2. These structures appear as repetitive peaks in the Kretschmann scalar as a function of proper time. However, due to the limited level of tuning and numerical resolution in the present simulations, these features cannot be conclusively identified as true (discrete) self-similarity. Achieving a more definitive verification would require finer tuning of the critical parameter and higher resolution to resolve the increasingly small spacetime scales involved.

But what makes tuning and classifying spacetimes difficult?

As the critical threshold is approached, it becomes increasingly difficult to classify a given evolution as subcritical or supercritical. In particular, near-critical evolutions may:

- not exhibiting a clear dispersal of curvature to spatial infinity,
- not form an apparent horizon detectable by standard horizon-finding tools,

thereby making the outcome ambiguous within finite numerical resolution. This could be attributed to several reasons.

Development of Coordinate Pathologies

Often, the inability to classify spacetimes arises from the development of coordinate pathologies, mainly constraint violations and coordinate singularities, in the evolution before we can place them into either supercritical or subcritical data.

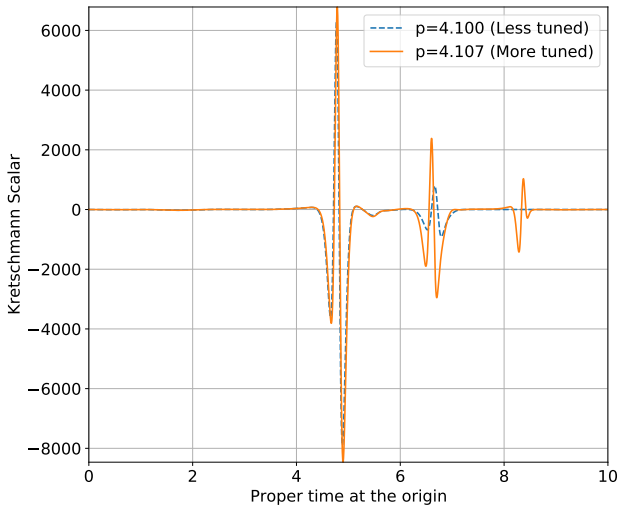


Figure 3.2: The figure depicts the Kretschmann scalar plotted against the proper time at the origin for the subcritical evolutions. Here, proper time refers to the time measured by an observer at rest at that spatial location in the spacetime. As expected in critical phenomenon, as we tune more from the subcritical side, we observe more echo-like features until they dissipate. Merely suggestive of self-similarity, further tuning and analysis are needed for conclusive identification.

Courant Condition and Resolution Limits

As we approach the critical regime, the solution develops structure on increasingly small spatio-temporal scales. This imposes a fundamental restriction on the timestep via the Courant–Friedrichs–Lewy (CFL) condition, which is required for the stability of explicit numerical schemes. For a hyperbolic system with characteristic speed $v_{\max}$, this leads to the condition

$$\Delta t \leq C \frac{\Delta x}{v_{\max}},$$

where $C < 1$ is the Courant factor determined by the numerical scheme and Δx and Δt are the spatial grid spacing and timestep spacing, respectively.

This condition has important consequences near criticality. To resolve the features, `BAMPS` uses Adaptive Mesh Refinement (AMR), which adjusts the spatial grid sizes to capture dynamics at smaller and smaller scales. As finer spatial resolution ($\Delta x \to 0$) is required to resolve the small-scale features that emerge, the timestep must decrease proportionally to satisfy the CFL condition. Consequently, the computational cost increases significantly, since the number of timesteps required scales inversely with Δt. With our current computational resources, we were only able to tune to about 3 decimal places.

Additionally, the dynamics of critical phenomena are such that, as we tune towards the threshold of collapse, the evolution resembles each other for an even longer duration, until the solution "chooses" an end point, be it a singular spacetime or dispersion. So essentially, we will have to evolve more simulation time as we tune the parameter more to reach the critical regime.

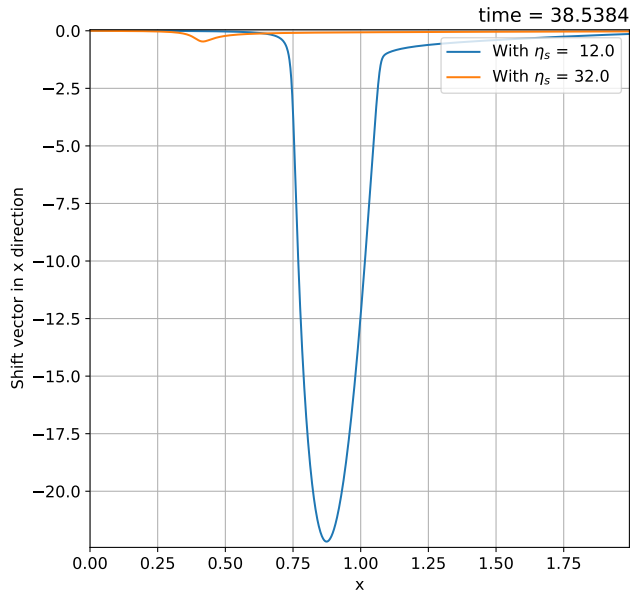


Figure 3.3: Two evolutions with different shift damping gauge parameters. As we can observe in the higher one, the divergence of the shift is damped away by the shift damping term (η_s) of the gauge, hence allowing us to evolve more in time and observe the dynamics.

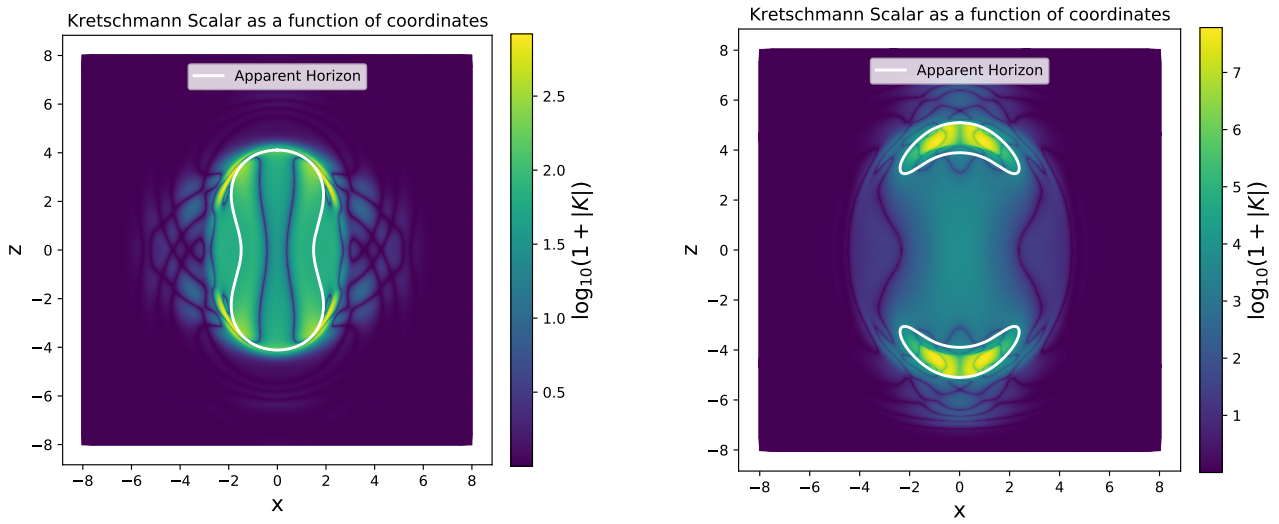
Therefore, the combination of shrinking physical scales and the CFL constraint severely limits how close one can tune to the critical parameter in multidimensional simulations. This is one of the primary reasons why achieving high precision in non-symmetric critical collapse is significantly more challenging than in spherical symmetry. This problem can most likely only be mitigated by evolving with more computational resources to effectively resolve the features in a way that doesn't require excessive computational time.

Gauge effects and their resolution

In near-critical evolutions, coordinate pathologies often manifest as the development of sharp features in the gauge variables, particularly in the shift vector and the lapse. One commonly observed issue is the formation of steep gradients or localized spikes in the shift, which can lead to numerical instability and eventual breakdown of the evolution.

These features are not physical but arise due to the choice of gauge conditions. In particular, certain gauge choices may fail to adequately account for the strong focusing of the geometry near criticality, leading to large coordinate distortions. Since, in NR, the coordinate system (indirectly) evolves alongside spacetime, the gauge dynamics can drive the coordinate system toward pathological configurations. In extreme cases, this can lead to the formation of coordinate singularities during the evolution. Such singularities are purely gauge effects and do not correspond to any physical divergence in the spacetime geometry, but they can nevertheless result in the breakdown of the numerical simulation.

To mitigate such coordinate pathologies, we increase the damping parameters in the HDW Gauge (Sec 2.2) to suppress rapid variations in the shift and control the growth of coordinate



(a) Peanut-shaped horizons

(b) Disjoint horizons

Figure 3.4: Plots here show the Kretschmann scalar over a reduced computational domain with the horizon found in the data overlaid on top. As we tune towards the threshold of collapse, we observe "peanut" (star-shaped) shaped horizons as in (a) found by the `AHLoc3D` and bifurcated or (b) disjoint "banana" (non-star-shaped) shaped horizons as we tune even more, which can only be found by the Cartesian-based AH finder.

distortions (See Fig.3.3). But finding the right gauge is a process that purely involves trial and error, and testing them requires evolving them with multiple gauge choices and understanding how the pathology behaves. Hence, this process takes time, which, in turn, slows down the tuning of the critical parameter.

Intricacies in Horizon Finding

Another cause of the inability to classify spacetimes stems from limitations in the AH finders we used. Initially, far away from criticality, we found circular (spherical) AH's, and as we approached the critical regime, we saw more of a "peanut" shaped horizons as in Fig.3.4a. These were comparatively found easily using the spherical parameterized `AHLoc3D` AH finder from the AH data outputted by `BAMPS`.

But as we tuned further, the inability to tune the parameter by finding horizons made us rethink the basic assumptions that we had made until now. (Other than the reasons mentioned before, like coordinate singularities and so on). As mentioned in Sec.2.3, we hypothesized that the horizon might already be present in the data, but we can't detect it because `AHLoc3D` uses spherical ray-body parameterization. This turned out to be true, as we were able to find horizons using a Cartesian parameterization-based AH finder [10], which was modified to account for twist.

As seen in Fig.3.5, the parameterization is no longer valid for horizons like these, and, as we enter the critical regime, we can only rely on this new parameterization. Currently, we find

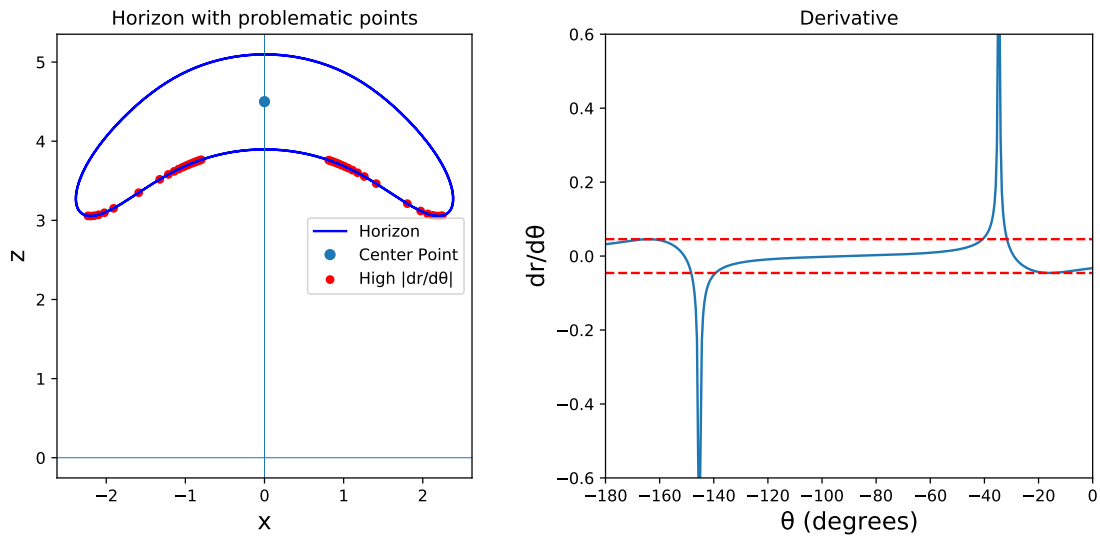


Figure 3.5: One of the disjoint horizons found by the new Cartesian-based horizon finder near criticality. `AHLoc3D` is not able to find them in a spherical parameterization, since the derivatives diverge near the problematic points shown in red if we were to find it using the $r = F(\theta)$ parameterization (borderline before multivaluedness).

disjoint horizons that are centered around $z = \pm z_{\text{peak}}$ where z_{peak} is the point on the z axis where the maximum of curvature (Kretschmann Scalar) is located.

Taken together, these results illustrate that while critical phenomena persist beyond spherical symmetry, accessing and characterizing them demands increasingly sophisticated numerical methods and a deeper understanding of the underlying geometry.

Chapter 4

Conclusion and Future Directions

In this work, we have investigated critical phenomena in axisymmetric vacuum spacetimes using Brill wave initial data with twist. By constructing and evolving one-parameter families of initial data, we identified a threshold separating dispersion from black hole formation. For the family of pure twisting Brill waves constructed here, the critical parameter has been tuned to a precision of three decimal places. While previous studies typically achieve higher precision (of order ~ 6 decimal places), the present level of tuning corresponds to an intermediate stage in the approach to the critical threshold. Nevertheless, it already allows us to observe a scaling relation between the maximum of the Kretschmann scalar and the distance from the critical parameter. An estimate of the critical exponent can also be obtained from this scaling behavior. However, given the current level of tuning, these results should be regarded as preliminary, and further refinement is required before drawing more definitive conclusions.

Additionally, near-critical evolutions exhibit echo-like features in the curvature, suggestive of underlying self-similar dynamics. While the current level of tuning and resolution does not allow for a definitive identification of discrete self-similarity, these results are consistent with expectations from critical collapse.

We have also discussed the numerical challenges encountered in this work and possible strategies to mitigate them. In particular, the use of a Cartesian-based apparent horizon finder has already improved the exploration of the parameter space and is expected to facilitate further tuning toward the critical regime. The substantial computational cost ($\approx$4000 core-hours per simulation) further highlights the difficulty of accessing the critical regime in less symmetric spacetimes.

Future Directions

An immediate direction for future work is to achieve higher precision in tuning the critical parameter. This would enable more accurate determination of critical exponents and a clearer assessment of universality. Exploring multiple families of initial data will be essential to establish whether the observed behavior is truly universal. Until now, we have only tried pure twisting of the Brill wave, i.e., the seed function $q(\rho, z) = 0$, and after trying different families in that, we will move on to adding the influence of the seed function along with twist variables. Then an even stronger test would be to take the other class of axisymmetric vacuum initial data, which are Teukolsky waves, modify them to include twist, and conduct similar tests in that data. We expect that to be easier with the insights that we have gained from this experiment.

A much deeper investigation of self-similarity is also required. Higher resolution simulations may

allow for the extraction of echoing structure and scaling periods, providing stronger evidence for the nature of the critical solution.

Understanding the role of twist remains an open problem. In particular, it is important to determine whether twist gives rise to new classes of critical solutions or modifies existing ones. This could provide insight into the broader structure of the solution space in axisymmetric vacuum spacetimes.

From a numerical perspective, improving gauge choices and developing more robust horizon-finding techniques will be crucial. We have already implemented a new gauge choice that appears to facilitate reaching the horizon more efficiently; however, it is not discussed here, as it requires further testing and validation. Additionally, the Cartesian-based apparent horizon finder currently requires substantial manual intervention. Developing automated or more robust horizon-finding methods will therefore be an important direction for future work.

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